# From target states to two common reports

One joint law; w=(3/5,2/5), a=(3/4,1/4), τ=1/5; M=6.

### A1. State 1: probability 3/4

| Unit | $\theta_1$ | $\theta_2$ | Composite c |
|---|---|---|---|
| **A** | 4 | 4 | 4 |
| **B** | 3 | 2 | 11/4 |
| **C** | 2 | 3 | 9/4 |
| **D** | 1 | 1 | 1 |

### A2. State 2: probability 1/4

| Unit | $\theta_1$ | $\theta_2$ | Composite c |
|---|---|---|---|
| **A** | 3 | 1 | 5/2 |
| **B** | 4 | 2 | 7/2 |
| **C** | 2 | 3 | 9/4 |
| **D** | 1 | 4 | 7/4 |

$c_i = (3/4)\theta_{i1} + (1/4)\theta_{i2}$

w weights errors; a combines target magnitudes.

### B. Forward error coefficients

| | AB | AC | AD | BC | BD | CD |
|---|---|---|---|---|---|---|
| Weighted direction | 1/4 | 1/10 | 1/10 | 2/5 | 1/10 | 1/10 |
| Composite | 1/4 | 0 | 0 | 0 | 0 | 0 |

### C. Weighted-direction optimum 1223

| A | B C | D |
|---|---|---|

g = (1,2,2,3); N = 5

| AB | AC | AD | BC | BD | CD |
|---|---|---|---|---|---|
| ● | ● | ● | ○ | ● | ● |

### C. Composite optimum 1123

| A B | C | D |
|---|---|---|

g = (1,1,2,3); N = 5

| AB | AC | AD | BC | BD | CD |
|---|---|---|---|---|---|
| ○ | ● | ● | ● | ● | ● |

Filled: forward direction. Open: withhold.

### D. Same N = 5; evaluate each loss

| Report | Weighted risk | Composite risk |
|---|---|---|
| 1223 | **17/120** | 3/40 |
| 1123 | 1/6 | **1/30** |

Both reports disclose five of six pairs.
Risk uses M=6; error fractions use N.
Compare minima within each target; loss magnitudes cannot choose the target.